Study setup and allocation

Research question: After using the simulator, can users correctly identify the effects of added monthly BNPL repayments and lower income on the monthly cash buffer? This is preparation only. No participant data have been collected.

Recruitment: three real people who did not develop the application. Project A Tech Team, PDF p.3, requires at least three test users and a structured feedback instrument. Marking Criteria A Tech Team, PDF p.4 (Results and Evaluation), also specifies at least three users and a structured instrument. Nondeveloper recruitment is an additional requirement of this protocol. Do not count developers or fictional QA as users.

Anonymous ID	Part 1: text only	Part 2: app
P01	Case A	Case B
P02	Case B	Case A
P03	Case A	Case B

Keep text first and app second for every participant. Only case allocation is alternated. With three participants the case distribution is 2:1, not fully balanced. Keep the unused case hidden until Part 2; collect or cover the first case sheet after its answers are locked.

Before anyone starts

Open https://bnplguard-yixuan502.streamlit.app/ without requiring participant login. Warm the service before timing, check Screen/Stress are usable, then start a fresh empty session. Use the same device, calculator policy and script within each session. Do not demonstrate answers or the default financial example beforehand.

Planned app revision: **771aadd**. Source: docs/deployment.md and outputs/deploy/stage8_v1/acceptance.json. Current GitHub main was checked as the same full revision. Live browser Screen loaded without a login prompt during material preparation. The page does not display a commit hash; version attribution combines the deployment receipt and unchanged main, not a browser-read hash. Recheck at each session; enter the actual verified commit in the blank workbook field. Stop and document an unexpected version or access failure.

Use the three preassigned IDs only. Do not record names, emails, ages or real finances. One member leads each session; an optional observer records notes silently. Keep one shared master workbook with a designated editor per session; do not overwrite another member's edits. Do not send the guide or workbook to participants.

Script and timing

Read this opening verbatim

"We are testing how this application communicates fictional monthly scenarios, not testing you. This takes about ten minutes. We will record anonymous answers, times and a short comment; no real financial details, names, audio or video. You may skip a question or stop. Please first work from one written case without the app, then use the app for a different case. A basic calculator is allowed in both parts. I will preserve your first answers before helping. Is that OK, and did you take part in developing this app?"

Record consent and nondeveloper confirmation in Sessions. If consent is declined or the person is a developer, do not run or score a session; do not fabricate a replacement record. Recruit another nondeveloper under a new explicit ID if needed.

Elapsed target	Action
0:00-1:00	Opening, consent, ID and case allocation.
1:00-3:00	Text-only case, six answers independently. Lock T1-T6.
3:00-6:00	Other case in Stress independently. Lock A1-A6.
6:00-7:00	Optional standardized assistance and revised answers. Keep originals unchanged.
7:00-8:30	Fictional Screen task and S1 interpretation. Preserve original before meaning help.
8:30-9:30	Subjective clarity and one verbatim comment.
9:30-10:00	Close and save. Debrief only after answers/ratings are locked.

Use a stopwatch. Start each active task when its case/instructions are available and stop at completion or the stated limit. Record seconds separately for text, app and assisted review. Pause only for observable technical loading/failure; record total paused seconds and reason. Do not pause for thinking, typing or navigation. Ten minutes is approximate; extend for technical delay and record it.

At each cutoff say: "Please give your independent responses now, including any you are unsure about." Copy exactly, mark unattempted items, then lock them. Do not correct or praise accuracy. If help is requested early, first lock the entire current part (including blanks); mark its independent phase ended, then help. Never describe assisted work as independent.

Assistance and recording

Before both independent case parts are locked, only repeat written wording verbatim or resolve access problems. If operational help is needed earlier, lock that part first. Defer numerical feedback and conceptual teaching until both independent parts are locked. Use the lowest applicable prompt and preserve its exact words.

Code	Permitted standardized prompt
H0	No help, or verbatim repetition only. Note which wording was repeated.
H1 navigation	"The four areas are at the top. Open Stress for the financial case."
H2 operation	"Use the field labels to enter the supplied case. Set currency to AUD, income reduction to 20%, and hypothetical BNPL reduction to zero."
H3 operation	"Use Compare scenarios. You can inspect a scenario and its displayed explanation."
H4 concept	Only after BOTH independent sets are locked: "Monthly cash buffer uses monthly income, living expenses and repayments. A change from current is a difference; the final buffer is a level. Please review your answer."
TECH	URL/load/connection assistance. Record pause time and cause. Do not supply numeric results.

Allow up to one minute of assisted review. Ask "Would you like to revise any answer?" Record a revised answer only if given. Do not overwrite originals. If no help/revision occurred, leave after-help response/score blank; do not copy an independent answer into it. Give no answer-key amounts until all responses and feedback are frozen.

Workbook entry rules

Sessions: one row per ID. Fill device/browser, actual app commit, consent and nondeveloper check, active seconds, paused seconds, whether any prompt occurred, subjective clarity 1-5, and one exact quote. Clarity and the quote are subjective, not comprehension scores.

Responses: one row per item, linked by ID and case allocation. T1-T6 = text Q1-Q6; A1-A6 = app Q1-Q6; S1 = Screen interpretation. Original answer and before score are separate from hint code/details, revised answer and after score. Item IDs are structural labels, not results. Copy the participant's actual words, including signs and units.

Before collection, blank means not yet collected. During a completed timed part, record "NR" for no answer and score 0. For withdrawal or technical failure, use "WITHDRAWN" or "TECH" and leave the score blank; retain the reason. Do not convert missing/failed sessions to zero. All populated scores must describe real observed responses.

Simulator-verified answer key

All amounts are AUD. Verified with the unchanged src.simulator_presentation.run_simulation in the project .venv, income_drop=0.2 and bnpl_reduction=0. The raw verification is retained in cases_verified.json beside these materials. These are standard answers, not participant observations.

Input	Case A	Case B
Income I	4,000	5,000
Living expenses E	2,200	2,800
Other debt D	600	700
Existing BNPL B0	300	400
New BNPL Bnew	200	300
Emergency savings S	800	1,200
Income reduction	20%	20%

Item in each part	Case A	Case B
Q1 current final buffer	900	1,100
Q2 new-BNPL final buffer	700	800
Q3 new-only change from current	-200 (decrease 200)	-300 (decrease 300)
Q4 combined FINAL buffer	-100	-200
Q5 combined CHANGE from current	-1,000 (decrease 1,000)	-1,300 (decrease 1,300)
Q6 combined classification	Shortfall	Shortfall

Key distinction: A combined final buffer is -100, while its change from the current 900 is -1,000. B combined final buffer is -200, while its change from the current 1,100 is -1,300. A reduction is not the remaining balance.

Current = I-E-D-B0. New only = current-Bnew. Combined = 0.8I-E-D-B0-Bnew. Income reduction is applied once to the stated baseline income. Each scenario starts from the current case; changes do not accumulate accidentally. Savings is a stock, not monthly income. E excludes repayments counted separately.

Scoring and interpretation

Objective cash-buffer scoring

Award 1 or 0 to each of Q1-Q6 using the key, identically in text and app parts. Q1/Q2/Q4: correct signed final amount. Q3/Q5: correct magnitude AND direction relative to current; "decrease 200" and "-200" are equivalent. A positive magnitude alone is ambiguous when direction is requested and scores 0. Q6: shortfall/deficit/negative buffer is correct. For a negative final amount, "100 shortfall" is equivalent to -100. Accept AUD/dollar notation and exact numeric equivalents within AUD0.01. No partial credit and no credit for method alone.

Compute text and app totals separately, each 0-6, from the before-help scores after collection. Keep assisted totals separately and only for items actually reviewed. Keep denominators explicit when items are TECH/WITHDRAWN. Do not treat blank uncollected cells as zero. Q6 records categorical understanding as well as Q4's numeric amount; these items are not independent observations.

S1: separate Screen interpretation, 0-2

One point: identifies the estimate as survey financial difficulty covering expenses/paying bills in a typical month. One point: states at least one valid scope limit, such as not BNPL default, not certainty about an individual, not future default/credit approval, or no Australian external validation. Do not give the first point to an answer identifying the target as BNPL default. Preserve contradictory wording for review. Do not add this score to the 0-6 simulator totals.

For this frozen fictional questionnaire, 83.85% is the displayed rounded example, not a required answer to S1. A suitable interpretation is an estimate of the study's contemporaneous self-reported financial-difficulty label. It does not establish a personal cause, a BNPL default probability or an approved borrowing amount. Record S1 before explaining this. If a meaning prompt is given afterwards, record the revised text and score separately.

After collection

A second member should review raw responses against the fixed key and resolve scoring differences in notes; do not change the key to fit results. Summarize actual item accuracy, within-person text/app totals, times, hints and quotes. Report subjective clarity separately. Three convenience users, the 2:1 case allocation, differing devices, case effects and the fixed text-then-app order limit inference. Practice and assistance can explain changes. This design cannot establish that the simulator causally improves understanding or represents a population. Do not invent feedback, significance or completed fixes.

Store completed materials privately in this project, under anonymous IDs. Keep names/contact lists out of the workbook and do not commit responses or quotes to the deployment repository. This round creates instruments only; no course report, deployment, model, threshold or financial formula is changed.